Assignment 4

Multimedia Technologies

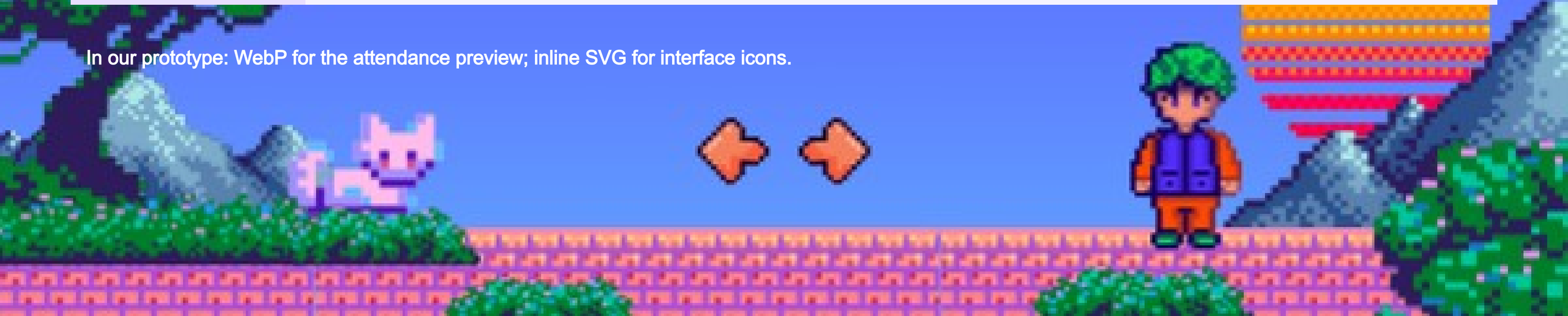
Team: Salima Zimbekkyzy, Azhar Saparzholova,
Nurassyl Oralbekov Group: SE-2623 Course: ICT



Part A — Multimedia formats

FORMAT	KEY TRAIT	BEST FIT
PNG	Lossless; supports transparency	Screenshots and graphics where crisp text or edges matter
JPEG	Lossy; compact for photographs	Photos and images where a smaller file is more important than perfect detail
WebP	Lossy or lossless; compact	Web images and the attendance preview used in our prototype

In our prototype: WebP for the attendance preview; inline SVG for interface icons.



A1. Format comparison

Format	Type	Compression	Alpha / motion	Typical size	Support	Project use
PNG	Image	Lossless	Alpha; APNG	Large photos	Broad	UI evidence
JPEG	Image	Lossy	Neither	Small photos	Broad	Photos
WebP	Image	Both modes	Alpha + motion	Compact	Modern	App preview
SVG	Image	Vector / text	Alpha + motion	Tiny icons	Broad	UI icons
MP3	Audio	Lossy	N/A	Small	Broad	Speech / music
AAC (M4A)	Audio	Lossy	N/A	Small	Broad*	Narration
WAV (PCM)	Audio	Uncompressed	N/A	Large	Broad*	Editing master
MP4 H.264	Video	Usually lossy	Motion; no alpha	Bitrate-based	Broad*	Demo delivery
WebM VP9	Video	Usually lossy	Motion; alpha*	Bitrate-based	Modern*	Web playback
MOV H.264	Video	Usually lossy	Motion; no alpha	Bitrate-based	Limited web	Editing / transfer

A container (MP4) packages tracks and metadata; a codec (H.264) encodes and decodes the media.

*Codec, OS and device support vary. File size depends on content, resolution, bitrate and duration.
Sources: MDN, Image file type guide and Media container formats.

A2. Measured experiment

Project file / export	Setting and dimensions	Size (KiB)	Quality evidence
attendance-preview.png	PNG, 390 × 844 px	58.39	Original; visual check pending
attendance-preview.jpg	JPEG Q80, 390 × 844 px	Not supplied	Export + inspect
attendance-preview-low.jpg	JPEG Q40, 390 × 844 px	Not supplied	Export + inspect
attendance-preview.webp	WebP Q80, 390 × 844 px	15.75	In app; visual check pending
10-second clip, version 1	Proposed: MP4 H.264, 720p	Not supplied	Record + export
10-second clip, version 2	Same clip, lower video bitrate	Not supplied	Compare text and motion

42.64 KiB saved per preview

58.39 → 15.75 KiB: 73.0% smaller at the same resolution.
Source: supplied project inventory, 23 Oct 2026. 1 KiB = 1024 bytes.
JPEG and clip rows are a test plan, not completed measurements.

A2. Visual quality check

What the quality comparison must show

Use the same attendance screen in all four exports. Compare an identical crop at 100% zoom so text edges and calendar lines remain comparable.

PNG

Image not supplied

JPEG Q80

Image not supplied

JPEG Q40

Image not supplied

WebP Q80

Image not supplied

Evidence still needed

One side-by-side screenshot, JPEG file sizes, and two exports of the same 10-second clip. Rate each result as sharp, acceptable or poor after inspection.

The source image and recording were not included in the supplied files. These boxes mark missing evidence.

A2. Our format choices

73%

smaller preview file

Same pixels. Less data.

PNG 58.39 KiB
WebP 15.75 KiB

Measured file sizes only.
Network speed needs its own test.

For the My AITU / LMS bridge, we use WebP at quality 80 for the attendance preview. At the same 390 × 844 resolution, its file size falls from 58.39 KiB to 15.75 KiB, saving 42.64 KiB, or 73.0%. We retain PNG as the original evidence image because it preserves text and edges without lossy compression. Inline SVG suits interface icons because vectors stay sharp at different sizes and share the HTML response. CSS provides the login pulse without a separate video download. For the project demonstration, we plan MP4 with H.264 video and AAC audio for broad playback support. JPEG settings and the ten-second clip still need measured comparison before we can report their quality or size.

B1. Interface icons

#	Asset	Where used	Vector viewBox	Markup size
3	Book	Courses / navigation	SVG 300 × 300	740 B
4	Calendar	Tasks / attendance	SVG 300 × 300	1,162 B
5	Graduation	Grades	SVG 300 × 300	857 B
6	User	Account button	SVG 300 × 300	869 B
7	Menu	Navigation button	SVG 300 × 300	561 B
8	Close	Navigation controls	SVG 300 × 300	441 B
9	Down	Account menu	SVG 300 × 300	391 B
10	Sign-out	Account menu	SVG 300 × 300	906 B
11	Refresh	Reload	SVG 300 × 300	945 B
12	Search	Course search	SVG 300 × 300	669 B
13	Eye	Password visibility	SVG 300 × 300	935 B
14	Check	Attendance	SVG 300 × 300	366 B

All icons: Flaticon Uicons Regular Rounded. Free licence with visible footer attribution.
Inline symbols share the HTML response; these are source sizes, not separate requests.

B1. Media asset inventory

#	Asset	Where used	Format / dimensions	Size	Source / licence
15	Absent icon	Attendance	SVG 300 × 300	442 B	Flaticon[1]
16	Clock icon	Attendance time	SVG 300 × 300	777 B	Flaticon[1]
17	Minus icon	Unmarked attendance	SVG 300 × 300	284 B	Flaticon[1]
18	Left arrow icon	Previous month	SVG 300 × 300	416 B	Flaticon[1]
19	Next arrow icon	Next month	SVG 300 × 300	404 B	Flaticon[1]
20	List icon	Attendance status	SVG 300 × 300	1,010 B	Flaticon[1]
1	Attendance preview	Sign-in	WebP 390 × 844	15.75 KiB	Own + icons[1]
2	LMS wordmark	Sign-in	CSS / text, 68 × 68	Embedded	Own
21	Login pulse	Login progress	CSS, 2 s loop	Embedded	Own
22	Entrance motion	Transitions	CSS, 0.4 s	Embedded	Own
23	Campus map	Map section	Responsive iframe	Variable	Yuujiso / MIT[2]
24	Inter font	Interface text	Web font / weights	Variable	Inter / OFL[3]
25	Remote view	Optional support	JPEG Q70, 1100 × 800	Variable	Private session

[1] Flaticon attribution in footer. [2] github.com/Yuujiso/aitumap (MIT), author beside map.
[3] github.com/rsms/inter (SIL OFL 1.1). Remote session views must stay out of the submitted demo.
Unused ui-right / ui-stats and evidence PNG files are not loaded by the running interface.

B2. Integration and optimisation

What is integrated

WebP preview: 390 × 844 px, displayed at 250 px wide.
18 inline SVG icons and a CSS login pulse.
Reduced-motion preference disables animation.
The full page still needs a 360 px browser check.

Controlled local payload	Before: PNG	After: WebP	Change
Preview image	59,787 B	16,126 B	−73.0%
4 front-end files + preview	141,601 B	97,941 B	−30.8%

DevTools measurements: still required

Disable cache, select Fast 4G, reload the same screen before and after. Capture transferred bytes, request count and Finish/Load time. Image-request delta should be 0; verify the total.

The payload calculation excludes headers, compression, API, fonts and map traffic. Stage 1 requires <3 s per request at 1,000 users; this remains unverified.

B3. Accessibility of media

Meaningful images have descriptive alt text

The test attendance preview describes the month and status counts. Decorative SVGs use aria-hidden="true"; controls retain text or accessible labels.

```
<img src="/attendance-preview.webp" width="390" height="844"
alt="Тестовый экран посещаемости: календарь за октябрь 2026 года
и счётчики — 0 присутствий, 1 пропуск, 0 опозданий,
1 запись без отметки.">
```

Motion follows the user’s preference

```
@media (prefers-reduced-motion: reduce) {
*, *::before, *::after { animation: none !important;
transition: none !important; scroll-behavior: auto !important; } }
```

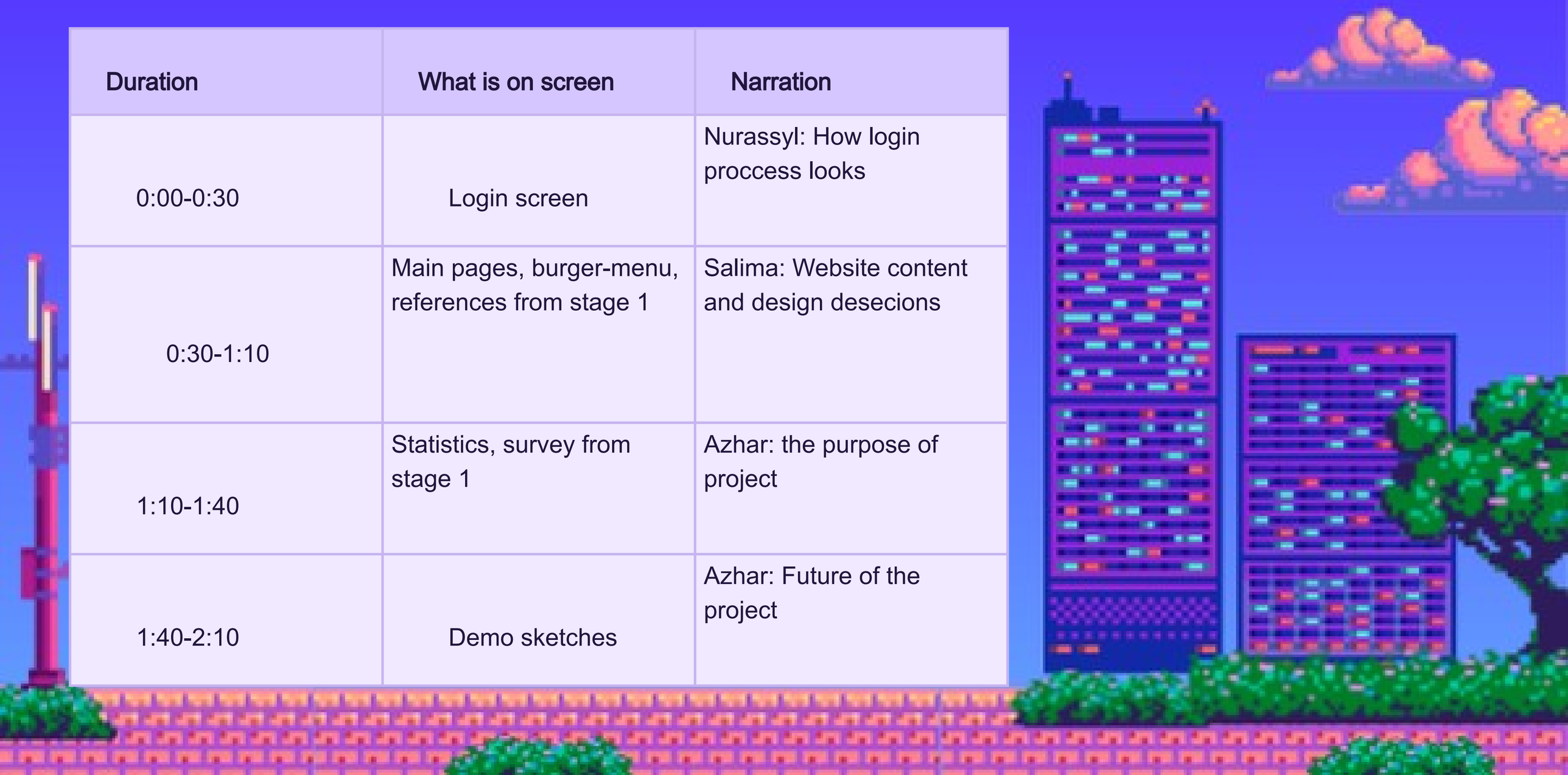
No speech media or sound playback: captions and sound equivalents are currently N/A. Progress is visible as text and a pulse. No live text sits over raster images, so the 4.5:1 text-over-image test is N/A. Source: supplied part_b.md, B3.

Part C — Multimedia presentation



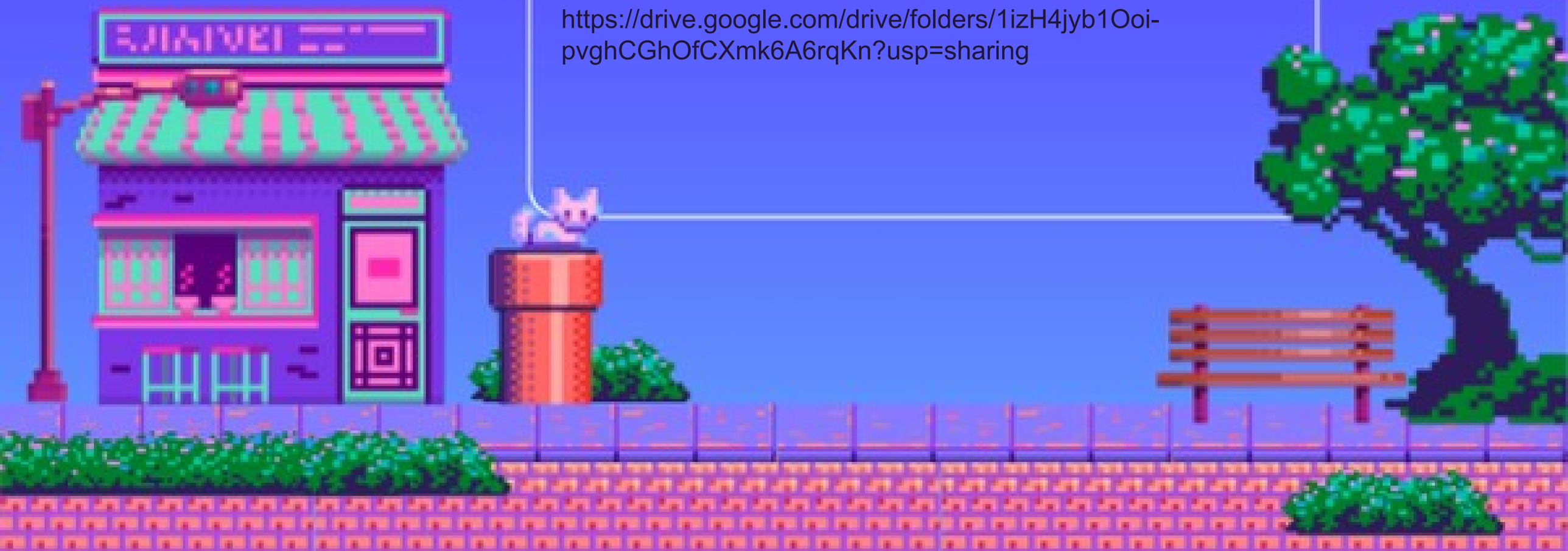
C1. Storyboard

Duration	What is on screen	Narration
0:00-0:30	Login screen	Nurassyl: How login proccess looks
0:30-1:10	Main pages, burger-menu, references from stage 1	Salima: Website content and design desecions
1:10-1:40	Statistics, survey from stage 1	Azhar: the purpose of project
1:40-2:10	Demo sketches	Azhar: Future of the project



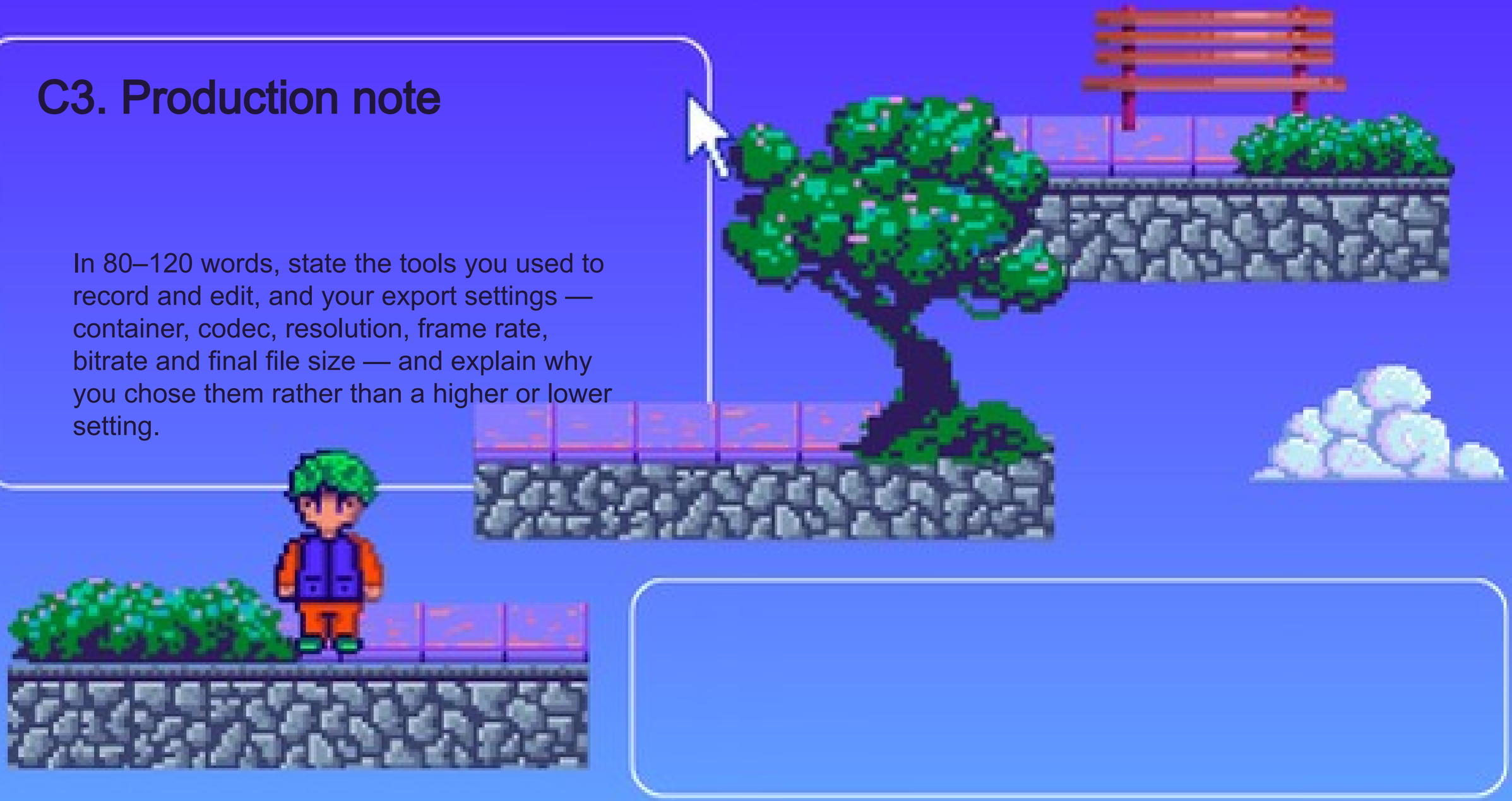
C2. The video

https://drive.google.com/drive/folders/1izH4jyb1Ooi-pvghCGhOfCXmk6A6rqKn?usp=sharing



C3. Production note

In 80–120 words, state the tools you used to record and edit, and your export settings — container, codec, resolution, frame rate, bitrate and final file size — and explain why you chose them rather than a higher or lower setting.



Part D — Integration with the project



D1. Updated prototype

https://github.com/d30mantcoder/ICTProject

Git clone it Create virtual environment Install dependencies (requirements.txt) Run bridge-test/app.py via python



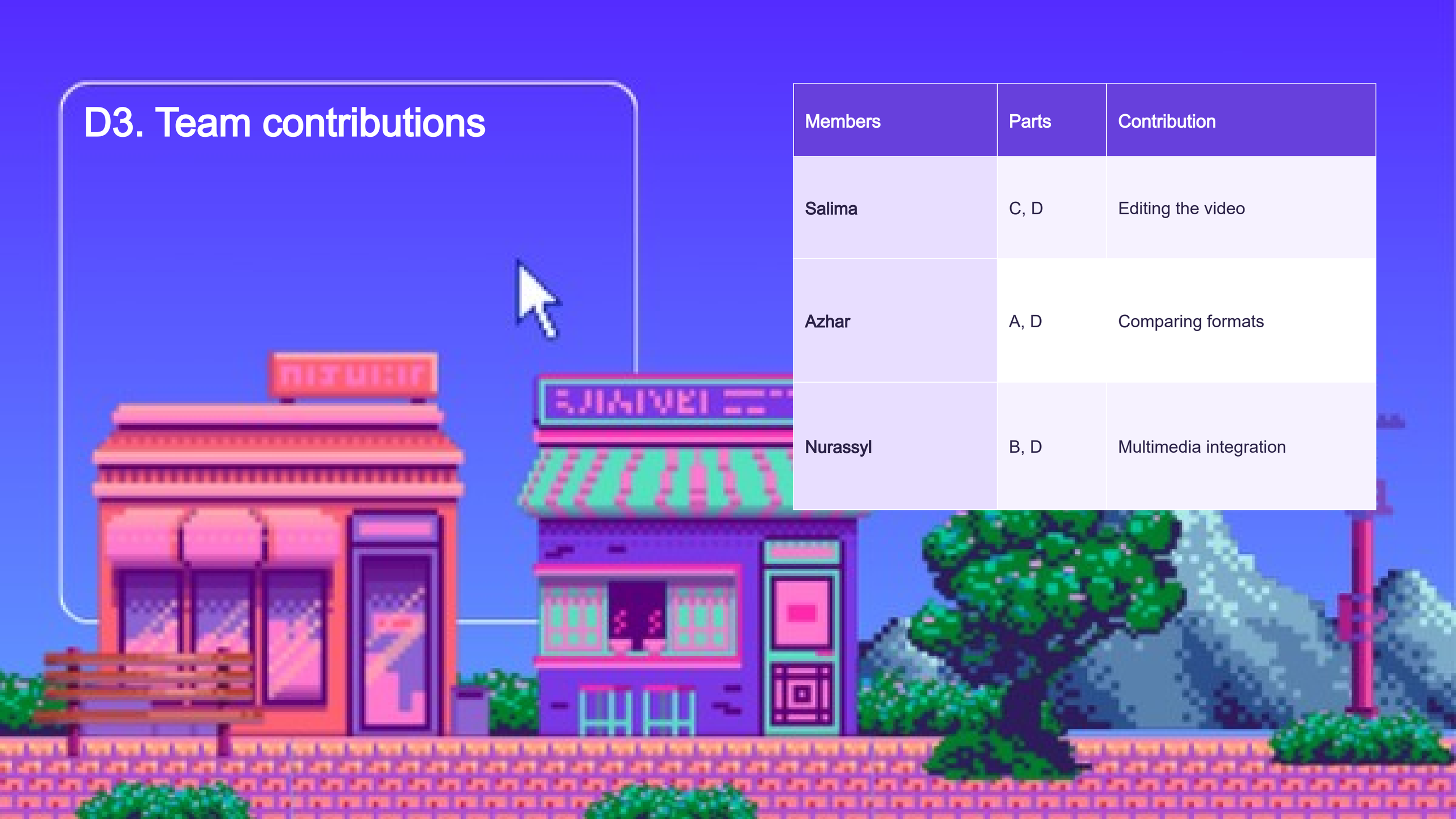
D2. Change log

Change	Reason	Status
Added fetching from LMS	To make the website usable	Using real LMS data
Redesign of UX of LMS	UX of original LMS was uncomfortable	Most useful buttons takes place on the main page
Added asynchronius sessions	It was impossible to work for more than 1 person at the same time before	Server handles multi-user sessions



D3. Team contributions

Members	Parts	Contribution
Salima	C, D	Editing the video
Azhar	A, D	Comparing formats
Nurassyl	B, D	Multimedia integration



Thank You

